Chapter 0 : Introduction to Statistics :-

Q: What is Statistics?

Ans: Statistics is the science of:

- Collecting (gathering) information.
- Presenting it (using, table, charts, etc).
- Analyzing data (with mathematical and graphical tools.
- Drawing conclusions (inference)

It uses mathematics and computers to help understand patterns in data and make decisions.

Q Why is Statistics Important?

- Helps us estimate population parameters from small samples.
- Used in hypothesis testing (a method to test assumptions).
- Crucial for:
  → Business
  → Market Research
  → Planning
  → Forecasting
  → Identifying patterns in data.

- **Conclusion:** Statistics is not just about numbers. It's about understanding, interpreting, and making decisions using data. Its a core foundation for becoming a good software engineer, researcher or data scientist.

Scanned with CamScanner

From Sample to Population :-

- Often we can't study an entire population, so we take a sample.
- Then we analyze the sample data to make conclusions about the full population.

• Role of Technology in Statistics :-

- With increasing data usage, data handling skills are in high demand.
- Statistics helps us understand real-world data we see every data.
- Leads to the emergence of new field called Data Science

Q. What is Data Science?

"A concept to unify statistics, data analysis, informatics, and related methods to understand and analyze actual phenomena with data".

★ Combines:
- Statistics
- Computer Science
- Mathematics
- Information Science
- Domain Knowledge

★ Data Scientist = Someone who:
- Write Code
- Applied statistical techniques
- Creates insights from data.

Scanned with CamScanner

Foundation of Statistics :-

- **Definition :-**

Statistics is the science of collecting, organizing, summarizing, analyzing, and interpreting data in order to make decisions or conclusions.

- **5 Main Steps of Statistics :-**

1) **Collecting** : Gathering raw data (e.g. surveys, experiments).

2) **Organizing** : Arranging data in tables or categories.

3) **Summarizing** : Using graphs/charts/measures (mean, median, etc)

4) **Analyzing** : Using formulas, comparisons, patterns.

5) **Interpreting** : Making decisions or conclusions based on data.

- **Real Life Example** : A company wants to know if a new mobile phone is liked by customers.
They do the following :

- Survey 500 people → collecting data.
- Put answers in a table → organizing
- Make a bar of yes/no responses → summarizing
- Find that 78% like it → analyzing
- Decide to launch it → interpreting.

Scanned with CamScanner

Types of Statistics :-

1) **Descriptive Statistics :** Describes the data you already have. (mean, charts, tables).

2) **Inferential Statistics :** Makes predictions or generalizations about a population from a sample.

- **Descriptive Statistics:**

- Focus on ~~the~~ what the data tell us directly.
- Does not predict or guess.
- Example: The average score in the class is 82.

- **Inferential Statistics:**

- Use a small group (sample) to say something about a larger group (population).
- Always includes probability.
- Example: "Based on sample of 100 voters, we estimate that 62% of the whole city supports candidate A."

* **Note :-**

Inferential stats always include a chance of being wrong!
You give a confidence level (like 95%) and mention the margin of error.

Scanned with CamScanner

Key Definitions And Terms :-

- **Data** : Raw facts or observations (e.g., height = 5.7 ft)
- **Population** : Entire group you want to study. (e.g., all students in Pakistan.)
- **Sample** : Small part of the population to study (e.g., 100 students).
- **Variable** : A property or feature that can change. (e.g. age, score, weight).
- **Parameter** : A value that describes the population (usually unknown).
- **Statistic** : A value that describes a sample (we calculate this)

* Example :-

Lets say :

- Population = All UBIT students
- Sample = 50 students from UBIT.
- Variable = Their GPA.
- Statistics = Average GPA of those 50 = 3.6
- Parameter = True average GPA of all UBIT students (we are estimating it).

Scanned with CamScanner

• Branches of Statistics :-

- Applied Statistics : Uses stats in real world fields.
- Mathematical Statistics : Focuses on theory behind stats.

• Types of Data (Intro) :-

- Qualitive (Categorical) : Non numeric, categories.
  (e.g., Colors, Brands, Blood group)
- Quantitive (Numerical) : Numbers you can calculate on.
  (e.g., Age, salary, weight)

• Role of Probability in Statistics :-

- Probability is the heart of inferential statistics.
- It tells us how likely an event is to happen.
- We will use it to :
  → Estimate population values.
  → Test hypothesis
  → Understand distributions (like binomial or normal).

★ Data Science = Statistics + Programming + Domain Knowledge

Scanned with CamScanner

Data
Primary Secondary (after operations)

Types of Data :-

A) Qualitive Data :-

This type of data cannot be measured numerically, but can be grouped or labeled.

Two Types :

- Nominal : No natural order or ranking (Blood group, gender, religion).
- Ordinal : Has meaningful order or rank (small, medium, large, Ratings).

B) Quantitive Data :-

This type of data can be measured and used in calculations.

Two Types :

- Discreate : Whole number counts (no fractions).
  Example : no of students, cars.
- Continuous : Any value within a range (includes decimals).
  Example : Height, Weight, Temperature.

Scanned with CamScanner

• Key Characteristics of Data :-

Characteristics	Description
Discreate vs Continuous	Discreate = countable, Continous = measurable
Grouped vs Individual	Grouped = in classes, Individual = raw values.
Scale of Measurement	Tells you how to treat and analyze the data.

• Scales of Measurement :-

The scale of measurement tell us :

- What type of data we are dealing with.
- Which statistical methods we can apply.
- Whether we can add, subtract, compare, etc.

There are 4 scales, from lowest to highest level:

1) ~~Normal~~ Nominal Scale :-

"Naming or labelling" scale, no order.

- Categories only
- You can count how many are in each group.
- No math can be done.

- Example : Favourite fruit

2) Ordinal Scale :-

"Order matters but differences don't".

- Categories are ranked or ordered.
- But you can't measure how much better one is than another.

Example : Star Ratings (★★★★★ Excellent, ★ bad)

* You know 5 stars > 1 star, but cant say its 2 Times better.

3) Interval Scale :-

"Ordered with equal ~~scal~~ spacing, but not true zero."

- You can do addition and subtraction.
- No multiplication / division (no true zero).
- Example : Temperature (°C)

$20°C - 10°C = 10°C$

but $20°C$ is not "twice as hot" as $10°C$.

Nominal = Name only
Ordinal = Order matters
Interval = Equal intervals but not 0.
Ratio = All math works, real zero.

4) Ratio Scale :-

"All math works, has true zero".

- Has a real zero (means "nothing")
- You can add, subtract, multiply, divide.

Examples: Age, Salary, Height.

★ Why this is so Important :-

- Helps you know which formulas graphs you can use.
- Makes sure you dont apply wrong method (e.g., you can't average "apple + mango".
- Forms the foundation of frequency tables, central tendency, and later probability.

Q: Identify the scale of measurement:

1) Shirt sizes: S, M, L →
2) Distance from home to university →
3) Temperature in Celsius →
4) Number of Programming Languages you know →
5) Blood Types: A, B, AB, O →

Scanned with CamScanner

Grouping Data & Frequency Distribution Tables :-

1. What is Grouping ?

Grouping means converting individual data values into class intervals (ranges).

- **Why Group Data ?**
  - Large raw data is hard to analyze directly.
  - Grouped data helps summarize and visualize the distribution.
  - Required for constructing frequency tables, histograms etc.

2. Frequency Distribution Table ?

A frequency table shows how many data values fall into each class interval.

- **Sturges Rule** - to find no of Classes :-

Sturges Rule Formula: $m = 1 + 3.322 \log_{10}(n)$

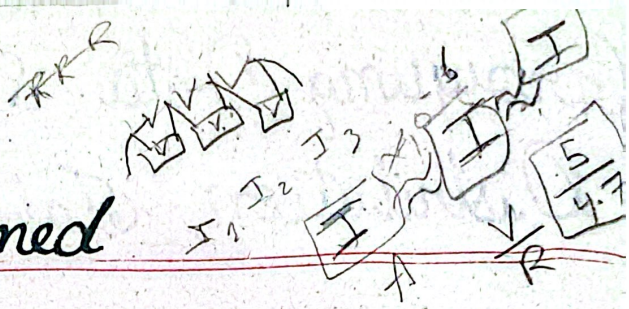
Where :

- m = number of classes
- n = number of observations
- Use log on calculator for base 10.

Always round up the value of m.

Scanned with CamScanner

Key Terms Explained



- **Class Interval**: A range of values (50–59, 60–69).
- **Class Width**: The size of each class.
- **Class Limits**: Lowest and highest value of a class. (e.g., 60–69 → lower = 60, upper = 69).
- **Class Boundaries**: Values that separate classes (adjusted to avoid gaps) eg., (59.5 – 69.5).
- **Midpoint**: Center of a class: (Lower limit + upper L) / 2.
- **Foreguency**: Number of data values in each class.
- **Comulative Freqr**: Sum of freqruencies from start to that class.
- **P(x)**: Relative freqruency = f ÷ total.
- **Steps to Construct a frequency table:-**

Lets assume the raw data values are: 45, 56, 67, 74, 83, 88, 91, 65, 72, 68, 78, 82, 69, 71, 74, 55, 62.

1. Find Min & Max:

- Min = 45
- Max = 91

Scanned with CamScanner

2. Use Struges Rule to Find Classes:

- $n = 17$
- $m = 1 + 3.332 \log_{10}(17) = 5.09 \Rightarrow$ **6 classes**

3. Find Range:

$$\text{Range} = \text{Max} - \text{Min} = 91 - 45 = \boxed{46}$$

4. Find Class Width: (Class Interval)

$$\text{Class Width} = \frac{\text{Range}}{\text{No of classes}} = \frac{46}{6} = 7.66 \Rightarrow \boxed{8}$$

- Always round up.

5. Create the Class Intervals:

Class Interval	Class Limits	Midpoint	Freqr	C.Freqr	Class Boundaries	P(x) = f/n
45 – 52	45 – 52	48.5	1	1	44.5 – 52.5	1/17 ≈ 0.06
53 – 60	53 – 60	56.5	2	3	52.5 – 60.5	2/17 ≈ 0.12
61 – 68	61 – 68	64.5	5	8	60.5 – 68.5	5/17 ≈ 0.29
69 – 76	69 – 76	72.5	4	12	68.5 – 76.5	4/17 ≈ 0.24
77 – 84	77 – 84	80.5	3	15	76.5 – 84.5	3/17 ≈ 0.18
85 – 92	85 – 92	88.5	2	17	84.5 – 92.5	2/17 ≈ 0.12

Scanned with CamScanner

• Final Frequency Distribution Table :-

Class Interval	Freq	Midpoint	C.Freq	P(x)
45 – 52	1	48.5	1	0.06
53 – 60	2	56.5	3	0.12
61 – 68	5	64.5	8	0.29
69 – 76	4	72.5	12	0.24
77 – 84	3	80.5	15	0.18
85 – 92	2	88.5	17	0.12

• Important Points :-

- Class Boundaries avoid gaps when plotting histograms.
- Midpoints are needed for mean, variance and graph center.
- Cumulative frequency is used for Ogives (cumulative graphs).
- P(x) is used in probability and expected value calculations.

• Relative Frequency :-

Relative Frequency tells us how often something happens compared to the total number of observations.

$$\text{Relative Frequency} = \frac{\text{Class Frequency}}{\text{Total Frequency}}$$

Scanned with CamScanner

Graphical Representation of Data :-

Q. Why use relative frequency ?

- Makes it easier to compare data sets with different sizes.
- Shows the proportion or percentage each class contributes to the whole.

- **Example :**

Lets say we have these class frequencies for test score.

Score Range	Frequency
40-49	3
50-59	7
60-69	10
70-79	5

Total Frequency = 3+7+10+5 = 25

Now calculate relative frequency.

Score Range	Frequency	Relative Frequency
40-49	3	3/25 = 0.12
50-59	7	7/25 = 0.28
60-69	10	10/25 = 0.40
70-79	5	5/25 = 0.20

- Relative frequencies can also be written as percentages:

0.12 → 12% , 0.28 → 28% , and so on.

Scanned with CamScanner

Graphical Representation of Data :-

Graphical representation of data: used to make data attractive, effective, ready for comparison and also save time and energy.

1. Bar Charts :-

Use Bar Charts when:
- Your data is in categories.
- You want to compare different things.

- **Definition**: A bar chart is a graph that uses rectangular bars to show the frequency or count of categorical or discrete data.

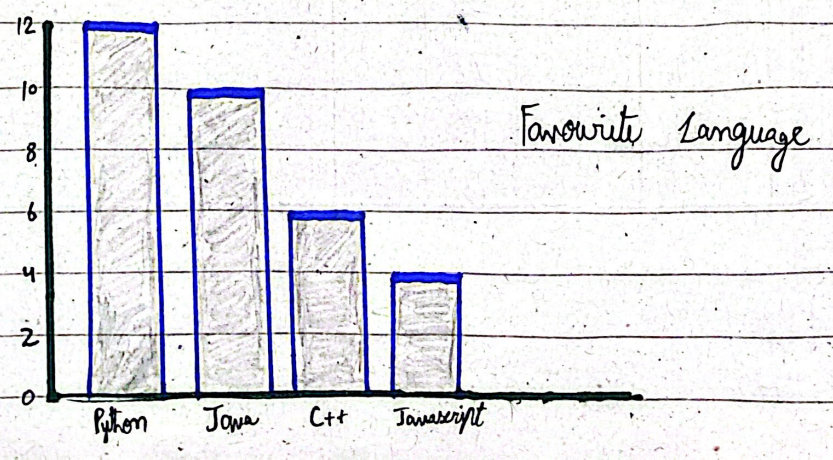
- **Key Features** :-

Bars should be of equal width.
Gaps between bars must visible.

Bars: Can be horizontal or vertical.
Gap between bars: Yes, because data is not continuous.
Data Type: Qualitives (nominal or ordinal) or discreate.
Axes: X-axis → Categories, Y-axis → Frequencies or Percentages.

- **Example**: Suppose you survey 30 students on their favourite programming languages.

Language	Frequency
Python	12
Java	8
C++	6
Javascript	4



Scanned with CamScanner

Use Histogram when:- Your data is in class intervals.
- You are working with grouped continuous data.

2) Histograms (For Continuous Data).

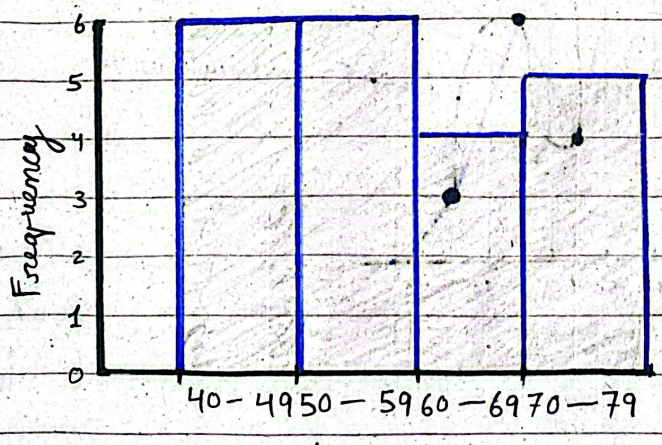
- **Definition**: A histogram is a graphical representation of a frequency distribution for continuous data using adjacent bars (no gaps).

- **Key Features**:-

Bars: Touch each other (because data is continuous).
Gaps between bars: No gaps.
Data Type: Quantitive - Continuous Data.
Axes: X-axis → Class Intervals, Y-axis → Frequency or Relative Freq

- **Example**: Suppose group student test scores.

Class Interval	Frequency
40–49	6
50–59	6
60–69	4
70–79	5



Histogram sketch

3. Frequency Polygon

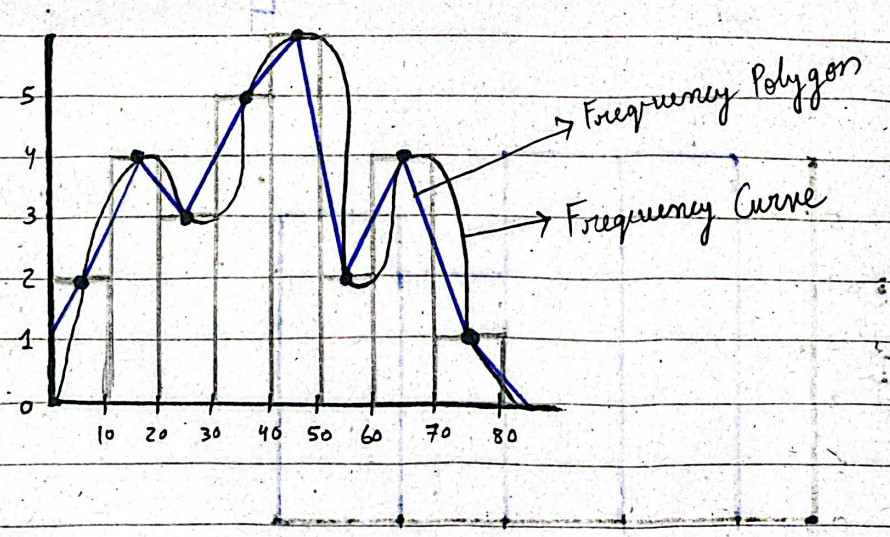
- **Definition**: A frequency polygon is a line graph that shows the shape of a frequency distribution. It connects the midpoints of each class interval with straight lines.

- **Features**:

Shape: Line graph (polygon)
Uses: Shows trend or distribution shape
Data Type: Quantitive (Grouped)
Points used: Midpoints of class intervals.
Touches X-axis: Yes, lines start and end at the X-axis.

• Steps to Draw a Frequency Polygon:

- Create a frequency distribution table.
- Find the midpoint of each class.
- Plot points (midpoint, frequency).
- Connect them using straight lines.
- Add two extra points on both end with freq = 0 to close.



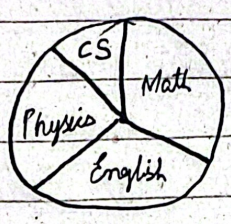
- Used to:
  - Compare multiple frequency distributions.
  - Visualize trends clearly.

Scanned with CamScanner

Pro Tip: Use different colors or shading for each slice.

4. Pie Chart

- **Definition**: A pie chart is a circular chart divided into sectors (slices) where each slice represents a proportion of the total.



- **Features**:

Shape: Circle.
Data Type: Categorical (Nominal/Ordinal)
Values represented by: Angles (each sector = % of 360°)
Comparison type: Proportional/percentage-based.

- **How to make a Pie Chart:**
  - Calculate the total frequency.
  - Find the percentage of each class:

$$\text{Percentage} = \left(\frac{\text{Class Frequency}}{\text{Total Frequency}}\right) \times 100$$

  - Convert to angle:

$$\text{Angle} = \left(\frac{\text{Class Frequency}}{\text{Total Frequency}}\right) \times 360^\circ$$

  - Draw a circle and divide it using the calculated angles.
  - Example:-

Subject	Students	Angle Calculation
Math	12	$12/30 \times 360^\circ = 144^\circ$
English	9	$9/30 \times 360^\circ = 108^\circ$
Physics	6	$6/30 \times 360^\circ = 72^\circ$
CS	3	$3/30 \times 360^\circ = 36^\circ$

  - All angles should add up to 360°.

Scanned with CamScanner

Measures of Central Tendency

"The average that best represents the data"

Q: What are Measures of Central Tendency ?

Ans They are numbers that represent the center or typical value in a data set.

- Mean : The "usual" average (sum ÷ count).
- Median : The middle value.
- Mode : The most frequent value.
- Weighted Mean : Mean with assigned weight (importance).
- Geometric Mean : Product-based average (used in growth rates, etc)
- Harmonic Mean : Inverse-based average (used in rates like speed).
- Truncated Mean :
- Square

1. Mean :-

- Definition : The mean is the sum of all values divided by the total number of values.

1. Mean :-

Definition: The mean is the sum of all values divided by the total number of values.

- **Formula :**

a) For ungrouped data:

$$\text{Mean } (\bar{x}) = \frac{\sum_i x_i}{n}$$

Where:

- $\sum_i x_i$ = sum of all data values.
- n = total number of values.

Class Boundary	Mid Point x_i	f_i	$f_i x_i$
1.45 – 1.95	1.7	2	3.4
	2.2	1	2.2
	2.2	4	9.8
	3.2	15	48.0
	3.7	10	3.7
	4.2	5	21
	4.9	3	14.1
		40	135.5

b) For grouped data:

$$\bar{x} = \frac{\sum_i f_i \cdot x_i}{\sum_i f_i} = \sum_i x_i P(x_i)$$

★ where:

- f = frequency
- x = midpoint of class.

- **Advantage**: Sensitive to any change in the value of any observation.
- **Disadvantage**: Very sensitive to outliers.

★ Example I: Data: 8, 4, 2, 6, 10

Solve

$$\bar{x} = \frac{\sum_{i=5}^{5} x_i}{5} = \frac{(8+4+2+6+10)}{5} = 6$$

Scanned with CamScanner

2. Median (Middle Value):-

Definition: The median is the middle value when data is arranged in ascending order.

- **Rules:**

- If n is odd:

$$\text{Median} = \text{middle value} = \boxed{\left(\frac{n+1}{2}\right)^{th} \text{ term}}$$

- If n is even:

$$\text{Median} = \boxed{\frac{\left(\frac{n}{2}\right)^{th} + \left(\frac{n+2}{2}\right)^{th} \text{ values}}{2}}$$

Example 1 ↔ Data: 2, 5, 7, 8, 10 → Median = 7 (3rd value)

Example 2 (Even Numbers) Data: 2, 5, 7, 8
→ Median = (5+7)/2 = 6

- **Median in Grouped Data (using Class Intervals):**

$$\boxed{\text{Median} = L + \left(\frac{\frac{n}{2} - C}{f}\right) \cdot h}$$

* Where:
  - L = lower boundary of median class.
  - n = total frequency.
  - C = cumulative frequency before median class.
  - f = frequency of median class.
  - h = class width.

Scanned with CamScanner

3. Mode (Most Frequent Value):-

Definition: The mode is the value that occurs most frequently in the data.

- Example (Ungrouped):

  Data: 2, 4, 4, 6, 8, 4
  → Mode = 4 (appears 3 times)

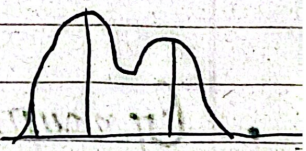
  1, 2, 3, 4, 5

  Mode does not exist.

- For Grouped Data:-

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

* where:
  - L = lower boundary of modal class
  - f_1 = frequency of modal class.
  - f_0 = frequency before modal class.
  - f_2 = frequency after modal class.
  - h = class width.



By Modal 2 Mode
TriModal 3 Mode

4. Weighed Mean:-

- Definition: A weighed mean gives more importance to some values based on their weights.

- Formula: $\bar{x}_w = \frac{\sum_i w \cdot x}{\sum_i w}$

where:
- x = values
- w = weights

Scanned with CamScanner

5. Geometric Mean (GM) :-

Definition: The Geometric Mean is the n^{th} root of the product of data values. Best for growth rates, percentages, ratios and multiplicative data

- ## Ungrouped Data Formula:

$$GM = \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot x_n} = \sqrt[n]{}$$

Or using logs:

$$\log(GM) = \frac{1}{n} \sum \log x_i$$

GM = antilog of result

ds	X	$\log x_i$
1		
2		
3		
4		
5		

- Example: Data: 4, 16, 64

→ $GM = \sqrt[3]{4 \cdot 16 \cdot 64} = \sqrt[3]{4096} = \boxed{16}$

- ## Grouped Data Formula:

$$\log(GM) = \frac{\sum f \cdot \log x}{\sum f}$$

GM = antilog of result

where x = class midpoint.

- Example

x (midpoint)	f	f. log(x)
10	2	2.000
20	3	3.903
30	5	7.385

$\sum f = 10$, $\log x = 13.288$

$\log(GM) = \frac{13.288}{10}$

$\log(GM) = 1.3288$

$\boxed{GM = 21.3}$

Scanned with CamScanner

6. Harmonic Mean (HM):-

Definition: The Harmonic Mean is used when the data involves rates, like speed, time, efficiency, etc.

• Ungrouped Data Formula:-

$$HM = \frac{n}{\sum_i 1/x_i} = \frac{1}{\bar{x}}$$

Where:

- n = number of values
- x_i = each individual value.

- Example: Speeds: 60 km/h, 40 km/h

$$HM = \frac{2}{\frac{1}{60} + \frac{1}{40}} = \frac{2}{\frac{5}{120}} = \boxed{48}$$

• Grouped Data Formula:-

$$HM = \frac{\sum f}{\sum (f/x)} = \frac{n}{\sum f_i/x_i}$$

where:

- f = frequency
- x = class midpoint (or data values)

Example:

x	f	f/x
10	4	0.4
20	3	0.15
40	3	0.075

$\sum f = 10$, $\sum (f/x) = 0.625$

$$HM = \frac{10}{0.625}$$

$$HM = \boxed{16}$$

Scanned with CamScanner

Q Which one is better: mean, mode or median

- The mean is valid only for interval data or ratio data.
- The median can be determined for ordinal data as well as interval and ratio data.
- The mode can be used with nominal, ordinal, interval, and ratio data.
- Mode is the only measure of central tendency that can be used with nominal data.

Measures of Dispersion :-

"How spread out the data is around the central value."

Q. What is Dispersion ?

Ans. Dispersion (also called variability) measures how much the data values deviate from the central tendency (like mean or median).

Q. Why do we study Dispersion ?

- Two data sets can have same average but different spreads.
- Helps us understand consistency or risk.
- Used in quality control, finance, machine learning etc.

• Types of Dispersion :-

- Absolute Dispersion : Actual spread in original units.
- Relative Dispersion : Spread as a ratio or percentage (unit - less).

• Measures We'll study :

Category	Measures
Absolute	Range, Quartile Deviation, Mean Deviation, Standard Deviation.
Relative	Coefficient of range, Quartile, Variation (CV), etc.

Scanned with CamScanner

Absolute Measure of Dispersion :- (A)

For measuring absolute dispersion there are 4 categories.

1. Range :-

Range is the simplest measure of dispersion. It tells us the difference between the highest and lowest values in a data set. OR "How wide the spread is - from minimum to maximum".

- Formula :- Range = Max value − Min value

$$R = X_{max} - X_{min} \quad [= X_m - X_o]$$

- For Grouped Data (class Intervals) : Range is still calculated the same way, but using upper boundary of last class and lower boundary of first class.

• Range Deviation :-

Range Deviation tells how far apart the highest and lowest values are on average, by taking half the range. It's the simplest measure of absolute dispersion, used for quick idea of spread.

- Formula :

$$\text{Range Deviation} = \frac{\text{Range}}{2} = \frac{X_{max} - X_{min}}{2}$$

• Notes :

$$\text{Group Data} = \frac{UCB^2 - LCB^2}{2}$$

- It's very basic and rough estimate.
- Highly sensitive to extreme values/outliers.
- Doesn't reflect what happens in the middle of the data.

Scanned with CamScanner

Q: What are Quartiles ?

Ans Quartiles divide the data into 4 equal parts:

- Q_1 = First Quartile (25% of data below)
- Q_2 = Second Quartile (= Median)
- Q_3 = Third Quartile (75% of data below)

2. Quartile Deviation (QD):-

Also known as:
Semi-Interquartile Range.

- Definition: Quartile Deviation is a measure of spread based on the middle 50% of the data. It calculates half the difference between the third and first quartiles. It ignores the extreme values (minimum and maximum), so its useful when the data has outliers of skewed.
- Formula: $$\text{Quartile Deviation (QD)} = \frac{Q_3 - Q_1}{2}$$

Q What are Quantiles ?

Ans Quantiles are the values that divide the data into equal parts.

- Quartiles: 4 equal parts
- Deciles: 10 equal parts
- Percentiles: 100 equal parts

Scanned with CamScanner

• Quartiles for Ungrouped Data :-

• Step-By-Step Method :

1) Arrange the data in ascending order.
2) Use positions to find :

- $Q_1 = \left(\frac{n+1}{4}\right)^{th}$ item OR $Q_i = \left(\frac{in}{4}\right)^{th}$ term
- $Q_3 = \left(\frac{3(n+1)}{4}\right)^{th}$ item. OR

3) Interpolate if needed.
4) Use formula for QD.

• Example (Ungrouped Data):

Data : 4, 7, 10, 12, 15, 18, 22, 24

Soln

$n = 8$

$Q_1 = \left(\frac{8+1}{4}\right) = 2.25^{th}$ item $\Rightarrow 7 + 0.25(10-7) = 7.75$

$Q_3 = \left(\frac{3(8+1)}{4}\right) = 6.75^{th}$ item $\Rightarrow 18 + 0.75(22-18) = 21$

$$QD = \frac{Q_3 - Q_1}{2}$$

$$QD = \frac{21 - 7.75}{2} = \boxed{6.625}$$

Scanned with CamScanner

Key Points:

- Use n for ungrouped N for grouped
- In group data always find class where position falls.
- Quartiles divide data into 4 parts.

• Quartiles for Grouped Data:-

- General Formula (Used for Q_1, Q_2, Q_3):

$$Q_k = XL + \left(\frac{kN}{4} - C\right) \cdot \frac{h}{f}$$

where:

- k = 1 for Q_1, 2 for Q_2, 3 for Q_3
- N = Total frequency (sum of all f)
- L = lower boundary of Quartile class.
- C = Cumulative frequency before quartile class.
- f = frequency of quartile class.
- h = class width.

- General Formula (For Q, D, P in Grouped Data):

$$X_k = L + \left(\frac{kN}{d} - C\right) \cdot \frac{h}{f}$$

X_k: $Q_1 - Q_3$, $D_1 - D_9$, $P_1 - P_{99}$

d: 4 for Quartiles, 10 for Deciles, 100 for Percentiles.

k: Which one you are finding ($Q_1 = 1$, $D_3 = 3$, $P_{75} = 75$).

Other variables L, N, C, f, h, same as before.

3. Mean Deviation (MD)

Also known as: Average Deviation

Q: What is Mean Deviation?

Ans Mean Deviation is the average of how far each value in the data is from a central value (mean or median), ignoring whether it is above or below. It uses the absolute value of deviation, so all differences are taken as positive.

Q: Why is it used?

- It tells us how spread out the values are.
- It gives a sense of consistency or variation.
- Unlike range or QD, it uses all values, not just a few.

- **Formula For Ungrouped Data:**

$$MD = \frac{1}{n} \sum |x_i - C|$$

where:

- x_i = each value
- C = central value (mean $\bar{x}$ or median)
- n = total number of values.
- Use absolute values: always positive.

Scanned with CamScanner

*) This is still not reliable solution because, we take absolute value changing negative into positive.

- **Formula for Grouped Data:**

$$MD = \frac{1}{N} \Sigma f_i |x_i - C|$$

Where:

- f = frequency
- x = midpoints of class intervals.
- $C = \bar{x}$ (usually the mean)
- $N = \Sigma f$

- **Step-by-Step: Mean Deviation (Ungrouped Data).**

1. Find the mean $\bar{x}$
2. Subtract mean from each value.
3. Take absolute value of each deviation.
4. Add them all.
5. Divide by n.

- **Step-by-Step: Mean Deviation (Grouped Data).**

1. Find midpoint x of each class.
2. Find $\bar{x} = \frac{\Sigma fx}{\Sigma f}$
3. Find $|x - \bar{x}|$
4. Multiply by f, get $f|x - \bar{x}|$
5. Sum of all $f|x - \bar{x}|$
6. Divide by $N = \Sigma f$.

Scanned with CamScanner

4. Standard Deviation (SD)

- **Definition:** Standard Deviation measures the average spread or deviation of values from the mean (center) of the data.
  It is most accurate and most widely used measure of dispersion.

- **Key Concept:**
  - The larger the standard deviation → the more spread out the data is.
  - The smaller the SD → the more consistent and closer values are two the mean.

- **Formula for Ungrouped Data :-**
  - Sample Standard Deviation:

$$s = \sqrt{\frac{\sum_i (x - \bar{x})^2}{n-1}} \quad \text{or} \quad s = \sqrt{\frac{\sum_i x^2 - (\sum_i x)^2/n}{n-1}}$$

  - Population Standard Deviation:

$$\sigma = \sqrt{\frac{\sum_i (x-\mu)^2}{N}} \quad \text{or} \quad \sigma = \sqrt{\frac{\sum_i x^2 - \mu^2}{N}}$$

  - μ (mu) : Population mean.
  - $\bar{x}$: Sample mean.
  - σ (sigma) : Standard Deviation of a population.

Scanned with CamScanner

- **Formula for Grouped Data :-**

- Sample Standard Deviation:

$$s = \sqrt{\frac{\sum_i f_i(x_i - \bar{x})^2}{N-1}} \quad \text{or} \quad s = \sqrt{\frac{\sum_i f_i x_i^2 - (\sum_i f_i x_i)^2/N}{N-1}}$$

- Population Standard Deviation:

$$\sigma = \sqrt{\frac{\sum_i f_i(x_i - \mu)^2}{N}} \quad \text{or} \quad \sigma = \sqrt{\frac{\sum_i f_i x_i^2}{N} - \mu^2}$$

- **Interpretation :-**

- $\sigma = 0 \rightarrow$ no variation, all values are the same.
- Higher $\sigma \rightarrow$ more variation (values are far from mean).
- Lower $\sigma \rightarrow$ more consistant and less spread.

5. Variance :-

Variance measures the average squared deviation of all values from the mean. It is the square of standard deviation.

Q Why do we use it ?

- It tells us how spread our data is.
- It's used in all further statistical formulas (including probability distributions, regression, etc).
- The unit of variance is square of the original unit. If the data is in kg, variance is in kg^2).

Scanned with CamScanner

10.9714
343.6714

• Ungrouped Data :-

• Population Variance :

$$\sigma^2 = \frac{\sum (x - \mu)^2}{n}$$

• Sample Variance :

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

• Grouped Data :-

• Population Variance :

$$\sigma^2 = \frac{\sum f_i x_i^2}{N} - \bar{x}^2 \qquad \sigma^2 = \frac{\sum f_i \cdot (x_i - \bar{x})^2}{N}$$

• Sample Variance :

$$s^2 = \frac{\sum f_i x_i^2 - (\sum f x)^2 / N}{N - 1}$$

* Note : You dont have to calculate variance separately if you already found standard deviation :

$$\text{Variance} = (\text{Standard Deviation})^2$$

Scanned with CamScanner

B) Relative Measures of Dispersion

Q: Why Relative Measures ?

Ans. Absolute measures (like range, SD, etc) depend on the units of data, so we cannot compare two different datasets directly.
That's where relative measures come in, they are unit-free ratios that help compare variability between different datasets, even if they have different units or scales.

1. Coefficient of Range :-

- Definition: The coefficient of range is a relative measure that compares the spread of data using the largest and smallest values only.
It is unitless and gives a sense of how widely data is spread.

- Formula :

$$\text{Coefficient of Range} = \frac{L - S}{L + S} = \frac{X_m - X_0}{X_m + X_0}$$

↓ Ungrouped Data

Where:
- L = Largest value (maximum)
- S = Smallest value (minimum)

$$C.R = \frac{UCL_{last} - LCL_{first}}{UCL_{last} + LCL_{first}} \times 100$$

→ Grouped Data

- **Used For:**
  - Quick comparisons of variability between datasets.
  - Works best when outliers are not extreme.
  - Simple but not very robust for detailed analysis.

- **Example:** Lets say for two datasets:
  - A: 12, 15, 18, 20 $\longrightarrow$ L = 20, S = 12
  - B: 2, 4, 5, 10 $\longrightarrow$ L = 10, S = 2

Coefficient of A: $\frac{20-12}{20+12} = \frac{8}{32} = \boxed{0.25}$

Coefficient for B: $\frac{10-2}{10+2} = \frac{8}{12} = \boxed{0.666}$

* B has more variability than A.

2. Coefficient of Quartile Deviation :-

- Definition: It measures relative spread based on the middle 50% of the data.

- Formula (Both Grouped and Ungrouped):

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$$

where:
- Q_1 = First Quartile (25% point)
- Q_3 = Third Quartile (75% point).

Scanned with CamScanner

3. Coefficient of Variation (C.V) :-

- Definition: It expresses standard deviation as a percentage of the mean. Helps compare variability between datasets of different scales.

- Formula:

$$C.V. = \frac{\sigma}{\bar{x}} \times 100$$

↓ population

$$C.V. = \frac{s}{\bar{x}} \times 100$$

↓ sample

where:

σ = Standard Deviation

$\bar{x}$ = mean

- **Comparison :-**

Measure	Range of Values	Interpretation
Coefficient of Range	0% - 100%	Higher → more spread
Coefficient of Q.D.	0% - 100%	Focuses on middle 50%
Coefficient of Variation	0% - ∞%	Lower → more consistency

Moments, Skewness & Kurtosis

1) Moments :-

Moments describe the characteristics (mean, spread, shape) of a distribution.

1.1 r^{th} Moment about any point 'a' :-

- Ungrouped Data :

$$\mu'_r(a) = \frac{1}{n} \Sigma_i (x_i - a)^r$$

- Grouped Data :

$$\mu'_r(a) = \frac{1}{N} \Sigma_i f_i (x_i - a)^r$$

1.2 Moments about origin (Raw Moments) :-

* set $a = 0$

- Ungrouped Data :

$$\mu'_r = \frac{1}{n} \Sigma_i x_i^r$$

- Grouped Data :

$$\mu'_r = \frac{1}{N} \Sigma_i f_i \cdot x_i^r$$

Scanned with CamScanner

1.3 Moments about mean (central moments)

* Set $a = \bar{x}$ = Mean.

- Ungrouped Data:

$$\mu_r = \frac{1}{n} \Sigma (x_i - \bar{x})^r$$

- Grouped Data:

$$\mu_r = \frac{1}{N} \Sigma f_i (x_i - \bar{x})^r$$

1.4 Relation Between Raw & Central Moments

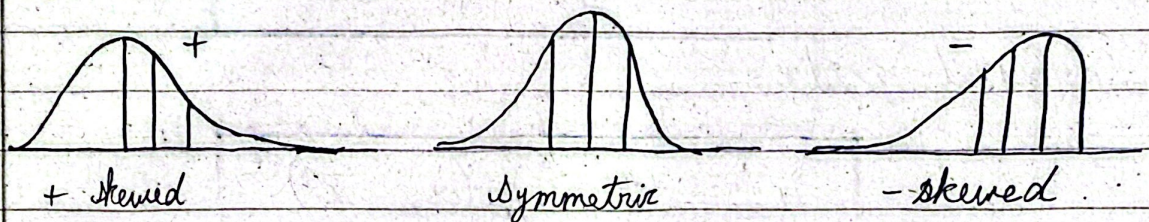
- $\mu_1 = 0$
- $\mu_2 = \mu_2' - (\mu_1')^2$
- $\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3$
- $\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$

Scanned with CamScanner

2) Skewness :- γ

Measures lack of symmetry.

- Positive skew : Long tail on the right side.
- Negative skew : Long tail on the left side.
- Symmetric : Tails equal.



- Formula :

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} \qquad \gamma = \sqrt{\beta_1}$$

- Look at μ_3 separately before squaring it.
  - If $\mu_3 > 0 \rightarrow$ Right (Positive) skew.
  - If $\mu_3 < 0 \rightarrow$ Left (Negative) skew.
  - If $\mu_3 = 0 \rightarrow$ Symmetric.

So basically :

- β_1 tells you the magnitude (how much skewness).
- Sign of μ_3 tells you the direction (left or right).

Scanned with CamScanner

3) Kurtosis :-

Kurtosis measures the peakedness or flatness of a frequency distribution compared to the normal distribution.

- Formula :.

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

where,

- μ_2 = variance
- μ_4 = 4^{th} central moment

- Interpretation of β_2.

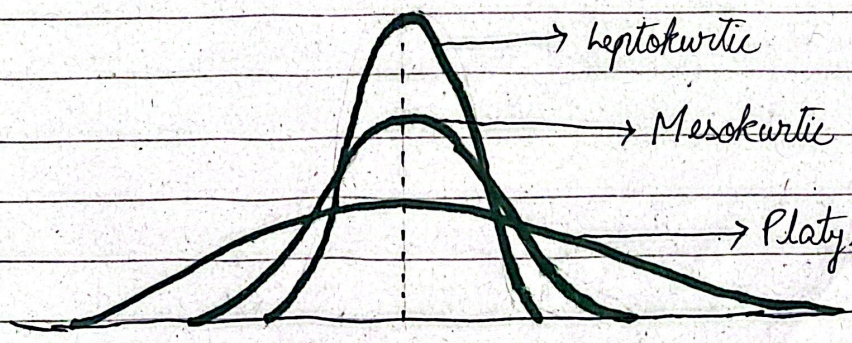
- $\beta_2 = 3 \rightarrow$ Mesokurtic
- $\beta_2 > 3 \rightarrow$ Leptokurtic
- $\beta_2 < 3 \rightarrow$ Platykurtic

Extra

Some book define excess kurtosis to compare with the normal distribution.

$$\gamma_2 = \beta_2 - 3$$

- $\gamma_2 = 0 \rightarrow$ Mesokurtic
- $\gamma_2 > 0 \rightarrow$ Leptokurtic
- $\gamma_2 < 0 \rightarrow$ Platykurtic



Scanned with CamScanner

Sheppard Correction

1) When data is grouped into class intervals, we assume each value in the class lies at the class midpoint.

- This introduce a rounding error in calculating moments.
- Sheppard's correction adjusts the moments to remove this error.

2) Formula for Correction :-

Let h = class interval width.

- Corrected 2nd central moment:

$$\mu_2(\text{corrected}) = \mu_2 - \frac{h^2}{12}$$

- Corrected 4th central moment:

$$\mu_4(\text{corrected}) = \mu_4 - \frac{h^2}{2}\mu_2 + \frac{7}{240}h^4$$

3) Why Needed :

- Without correction → variance & kurtosis become slightly larger than the actual values.
- Sheppard's correction → gives true moments when data is continuous but grouped.

Stem-and-Leaf Plot (Stem Plot):-

A stem-and-leaf plot is a method of organizing numerical data where:

- The stem represents the leading digit(s).
- The leaf represents the last digit.

It displays data values while maintaining the original numbers (unlike histograms where exact values are lost).

• Purpose:-

- To organize raw data into compact form.
- To see the distribution shape (spread, clusters, gaps).
- To quickly locate median, mode, and range.
- Useful for small to moderately sized data sets.

• Example:

- Data (exam scores): 43, 47, 52, 55, 59, 61, 62, 67, 70, 74

Stem	Leaf
4	3 7
5	2 5 9
6	1 2 7
7	0 4

→ Symmetric

Key: 4|3 = 43

- Organize the data in ascending order first.

Box Plot (Whisker Plot)

A box plot is a graphical summary of data based on the five-number summary.

1) Minimum
2) First Quartile (Q_1)
3) Median (Q_2)
4) Third Quartile (Q_3)
5) Maximum

Uses:
- Comparing two/more datasets.
- Quickly checking for skewness and outliers.
- Better for large data than stem and leaf.

• Purpose :-

- To show spread and symmetry.
- To identify outliers.
- To compare distributions between groups.

• Construction Steps :-

1) Arrange data in order.
2) Find :
   - Median (Q_2)
   - Q_1 (median of lower half)
   - Q_3 (median of upper half)
3) Draw a number line.
4) Draw a box from Q_1 to Q_3.
5) Draw a vertical line inside the box at Median (Q_2).
6) Draw whiskers from box to min and max (excluding outliers).
7) Plot outliers separately (dots or stars).

Scanned with CamScanner

* Whiskers lower inner fence and upper inner fence tk hi draw kr sakte hain.

SO = suspected Outliers

Example :-

Data : 11, 22, 20, 14, 29, 8, 35, 27, 13, 49, 10, 24, 17

1) Sort the data :

Min 8, 10, 11, 12 | 13, 14, 17, 20 (Q_2 Median), 22, 24, 27, 28 | 29, 35, 49 Max

(Q_1 = 12, Q_3 = 28)

$$\text{Median } (Q_2) = \left(\frac{n+1}{2}\right)^{th} = \frac{13+1}{2} = \frac{14}{2} = 7^{th} \text{ term} =$$

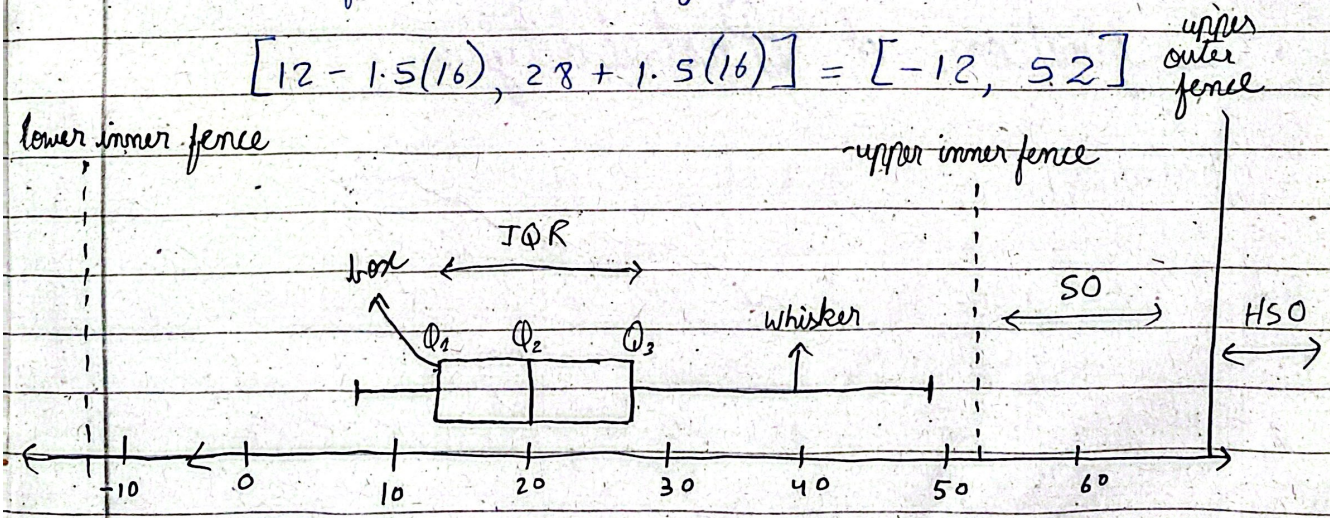
$$Q_1 = \left(\frac{1(13)}{4}\right)^{th} = 12 \qquad Q_3 = 28$$

$$IQR = Q_3 - Q_1 = 28 - 12 = 16$$

Range $[Q_1 - 1.5(IQR),\ Q_3 + 1.5(IQR)]$

↳ outlier if outside this range.

$$[12 - 1.5(16),\ 28 + 1.5(16)] = [-12,\ 52]$$



Range $[Q_1 - 3(IQR),\ Q_3 + 3(IQR)]$

↓ iske baher honge to highly suspected outlier (HSO).